

Typography, iconography & motion — type ramp, icon system, motion tokens

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Master technical specification — Volume 10 (Design System), nano-detail deep-dive. This document **owns** three of HelixVPN's design-token families that sit alongside the colour system: the **typographic ramp** (display / headline / title / body / label tiers, each a triple of size / line-height / weight, all as tokens), the **font-family strategy** per platform (system vs bundled, the §11.4.162 no-overlap / no-collision concern, font licensing flagged as a decision), the **iconography system** (icon-set strategy, the sizing grid, the per-platform asset forms SVG / icon-font / native), and the **motion / animation token system** (durations, easing curves, the named transitions that drive the connection-state UI, and the reduced-motion accessibility contract).

SPEC-ONLY. It defines *what the type/icon/motion tokens are and how they resolve into each platform's form* — not the shipping `helix_design` build. The token **structure** these live in (tiers, schema, the `$value/$type` envelope, the `font.* / motion.*` categories) is owned by `[design-tokens.md]`; the **colours** the type and icons are painted in by `[color-system.md]`; the **multi-form emit** (how a `font.semantic.body.md` becomes a SwiftUI Font, a Compose TextStyle, ...) by `[token-export-pipeline.md]`. This document is **original HelixVPN design work** — the ramp, the icon grid, the motion curves are owned here.

Boundary with sibling docs. Owns: the type-scale values, the font-family matrix, the icon grid + asset-form policy, the motion durations/curves/named transitions, the reduced-motion contract. Consumes: the token tier/naming model

- the font/motion \$type envelopes [design-tokens.md §2, §3, §6.4]; the connection-state palette + the 7-variant ffi::TunnelStatus the motion transitions are keyed to [color-system.md §3], [v04-client/ffi-surface.md §3.2]; the 3-app / 8-platform matrix [SPECIFICATION.md §3]; the OpenDesign DESIGN.md §3 (Typography) / §6 (Motion) authoring form [opendesign-foundation.md §3.2].

Evidence base. [DT §N] = final/v10-design/design-tokens.md; [COLOR §N] = final/v10-design/color-system.md; [OD §N] = final/v10-design/opendesign-foundation.md; [OV §N] = final/v10-design/00-overview-and-submodule.md; [FFI §N] = final/v04-client/ffi-surface.md; [SPINE §N] = final/SPECIFICATION.md. Claims not grounded in the evidence base or in this document's own original design choices are tagged UNVERIFIED per constitution §11.4.6 — never fabricated. Font *licensing* facts (Inter, JetBrains Mono) are tagged UNVERIFIED pending the verification pass mandated by §11.4.99 (latest-source cross-reference before any guide is published).

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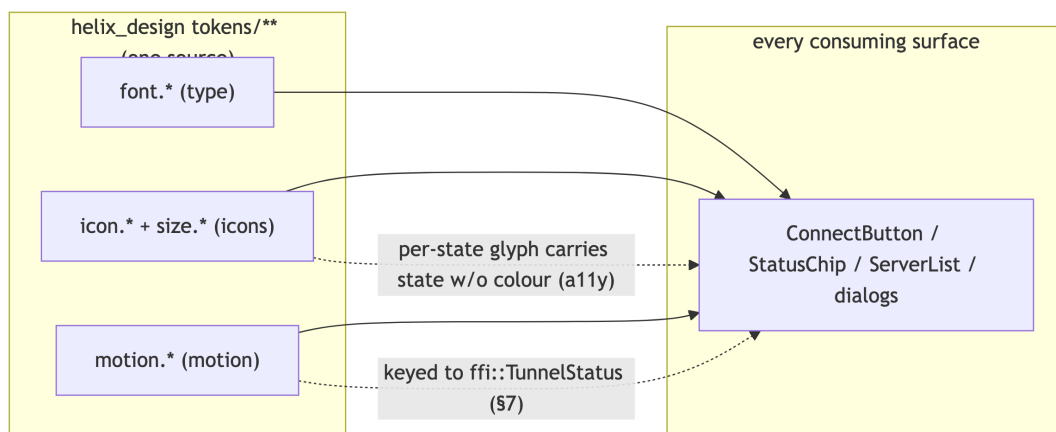
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0. Position & the three families

The colour system [color-system.md] answers *what colour means what*. This document answers the other three questions a design system must answer for every one of HelixVPN's **3 apps × 8 platforms**:

Family	Category root [DT §2]	Owns	The product-defining slice
Typography	font.*	the type ramp (size/line-height/weight tiers), font families, the per-platform font emit	the StatusChip / ConnectButton label legibility; mono for keys/IPs
Iconography	icon.* (+ size.* for the grid)	the icon set, the sizing grid, the asset forms (SVG / font / native)	the per-state icon (shield-check, spinner, warning-triangle) that carries state without colour (CVD a11y, [COLOR §0])
Motion	motion.*	durations, easing curves, named transitions, reduced-motion	the Connecting/ Reconnecting pulse, the state cross-fade, the connect ring

All three obey the same three-tier token model [DT §1] (primitive → semantic → component) and the same one-source / zero-drift rule [DT §0]: one JSON source, every consumable form generated [token-export-pipeline.md]. All three also obey §11.4.162: light + dark where colour is involved (icons are colour-tinted, so they ship a light + dark tint), no element overlap, no label overlay, and visual-regression coverage on every change.



1. The typographic ramp

The ramp is **five tiers** — display, headline, title, body, label — each with a small closed set of sizes. Every step is a **triple**: a font **size** (dimension), a **line-height** (number, unitless multiplier), and a **weight** (fontWeight). The ramp is a **modular scale** anchored at `body.md = 14px` with a ratio of **1.25** (major third) rounded to the 4-pt grid [DT §6.1] so every size lands on a clean device pixel.

1.1 The full ramp (all tiers, all steps)

Token	Size	Line-height (×)	Computed LH	Weight	Family	Typical use
<code>font.semantic.display.lg</code>	40px	1.10	44px	bold 700	sans	onboarding hero, splash
<code>font.semantic.display.md</code>	32px	1.15	37px	bold 700	sans	empty-state title
<code>font.semantic.headline.lg</code>	28px	1.20	34px	semibold 600	sans	screen title (Console master)
<code>font.semantic.headline.md</code>	24px	1.25	30px	semibold 600	sans	dialog title, section header
<code>font.semantic.title.lg</code>	20px	1.30	26px	semibold 600	sans	card title, ConnectBu label
<code>font.semantic.title.md</code>	18px	1.35	24px	medium 500	sans	list-section title
<code>font.semantic.title.sm</code>	16px	1.40	22px	medium 500	sans	ExitPicker row primary
<code>font.semantic.body.lg</code>	16px	1.50	24px	regular 400	sans	reading copy (settings header)
<code>font.semantic.body.md</code>	14px	1.50	21px	regular 400	sans	default body — most text
<code>font.semantic.body.sm</code>	13px	1.45	19px	regular 400	sans	secondary, supporting text
<code>font.semantic.label.lg</code>	14px	1.20	17px	medium 500	sans	button label, tab
<code>font.semantic.label.md</code>	12px	1.30	16px		sans	

Token	Size	Line-height (×)	Computed LH	Weight	Family	Typical use
				medium 500		StatusChi text, badge
font.semantic.label.sm	11px	1.30	14px	semibold 600	sans	underline, uppercase eyebrow
font.semantic.mono.md	13px	1.45	19px	regular 400	mono	IP / pubke fingerprint region cod
font.semantic.mono.sm	12px	1.40	17px	regular 400	mono	inline key fragments logs

Why anchor at body.md = 14px, not 16px. HelixVPN's densest surfaces are the Console server table and the Client ExitPicker/log — list-heavy, many rows visible at once. A 14px body keeps row density high while body.lg = 16px serves prose (settings help). Reading copy that the user *reads* (not scans) uses body.lg; chrome/labels use label.*. This is the original HelixVPN choice; both clear the platform minimum-legible floors (iOS Dynamic Type body 17pt is met by body.lg for accessibility-large; the 14px default is a dense-UI baseline, never the only size for body prose).

1.2 The modular scale (visualised)



1.3 The tier semantics (when to use which)

Tier	Role	Never used for
display.*	one-per-screen marketing/hero moment (onboarding, splash, empty-state)	running UI; it is too large to repeat
headline.*	the screen's or dialog's single title	body or labels
title.*	section / card / row titles; the ConnectButton label (title.lg, large-bold → governs AA-large [COLOR §3.3])	long prose
body.*	sentences the user reads or scans; settings descriptions	a button label (use label.*)

Tier	Role	Never used for
label.*	chrome: buttons, tabs, chips, badges, the StatusChip (label.md)	a paragraph
mono.*	anything the user might copy/compare character-by-character — IP, WireGuard pubkey, fingerprint, region code, log line	prose (proportional reads faster)

mono.* is product-load-bearing. A VPN UI shows keys, IPs, fingerprints and exit identifiers the user verifies character-by-character (is this *really* my pubkey?). A proportional font makes 1/1/I and 0/0 ambiguous — a real security-legibility hazard, not a stylistic preference. Every such value uses mono.* (a slashed-zero / disambiguated mono — see §2.3).

2. Font families per platform (system vs bundled)

2.1 The strategy — bundled sans/mono for brand consistency, with a system fallback chain

HelixVPN's primary surface is **one Flutter codebase** [SPINE §3], where bundling the font guarantees the **same** glyphs render on all 8 platforms (a system-font strategy would make iOS render SF Pro, Android Roboto, Windows Segoe — the "connected" label would have a different shape per OS, a brand-consistency leak). The chosen bundled families (primitive tokens [DT §4]):

Primitive token	Family	Role
font.primitive.family.sans	Inter	all proportional text (display→label)
font.primitive.family.mono	JetBrains Mono	all monospace (keys/IPs/logs)

These bind the semantic ramp's family column (§1.1). The **fallback chain** (if the bundled font fails to load, or on a native shim that cannot bundle) degrades to the platform system font, *never* to a random serif:

```
sans:  Inter → -apple-system / SF Pro (Apple) → Roboto (Android/Harmony) →
      Segoe UI (Windows) → system-ui (Linux/Web/Aurora) → sans-serif
mono:  JetBrains Mono → ui-monospace / SF Mono (Apple) → Roboto
```

Mono (Android) →
Cascadia Code / Consolas (Windows) → monospace

2.2 The per-platform decision matrix

Platform	Flutter app surface	Native-shim surface (small)	Font delivery
iOS / macOS	bundle Inter + JetBrains Mono in the Flutter asset bundle	NE-config-UI / widgets (SwiftUI) — system font (SF Pro / SF Mono), brand sans not bundled into the tiny extension	Flutter bundles; native shims use system per §2.4
Android / HarmonyOS	bundle in Flutter	quick-settings tile / VPN-ability (Compose / ArkTS) — system font (Roboto / HarmonyOS Sans)	as above
Windows / Linux	bundle in Flutter	—	Flutter bundles
Web (Console)	bundle as a @font-face web font (woff2), with the fallback chain in the CSS export	—	self-host woff2 (no third-party CDN — privacy, [COLOR \$0])
Aurora	bundle in Flutter; Qt/C++ shim (C-Qt export) uses the system font	Qt surface — system font	as above

D-TYPE-1 (surfaced, §10). *Native shims use the system font, not the brand font.* The iOS NE-extension / Android tile / HarmonyOS ability / Aurora Qt surfaces are tiny config UIs; bundling a 300 KB+ font family into each is disproportionate, and these surfaces show little text (a toggle + a status line). They render the **same palette** (the colour token export reaches them, [OV §6]) but the **system font** — a deliberate, documented exception to brand sans, justified by binary-size + the surfaces' minimal text. The Flutter app — where the user spends ~all their time — is always brand-font.

2.3 Mono disambiguation requirement

The chosen mono (`font.primitive.family.mono`) **MUST** disambiguate the security-critical glyph pairs 0/0, 1/l/I, 5/S, 8/B. JetBrains Mono ships a slashed/dotted zero and distinct 1/l/I by design — the reason it is chosen over a generic mono. The §9 gate CM-type-mono-disambiguated is a *manual* design-review checkpoint (an automated glyph-shape oracle is UNVERIFIED — flagged §11.4.6, not claimed).

2.4 Licensing — flagged as a decision (§11.4.99)

D-TYPE-2 (open, §10) — font licensing. Inter (SIL Open Font License 1.1) and JetBrains Mono (Apache-2.0 / OFL depending on distribution) are both understood to permit embedding + web self-hosting, **but** the exact license text, version, and bundling terms **MUST** be verified against the **latest** upstream source per §11.4.99 (latest-source doc cross-reference) **before** any font file is committed to `helix_design/assets/fonts/` or a setup guide is published. The license file + the verified URL + access date go into `assets/fonts/<family>/LICENSE` and the §11.4.99 "Sources verified" footer. Marked **UNVERIFIED** here: the OFL/Apache assumption is *not* asserted as fact — it is the hypothesis the verification pass confirms. If a license turns out to forbid embedding, the fallback is a system-font-only strategy (§2.1 fallback chain becomes primary) — a §11.4.66 operator decision, not a silent swap. Composes [OV §13 D-DESIGN-5] (bundle-vs-reference font decision).

3. Type tokens — the JSON shape & per-platform emit

3.1 The composite type token

A type-ramp step is a **composite** — a small object of `size / lineHeight / weight / family` references, each a leaf of its own `$type` [DT §3.1]. The ramp lives in the semantic tier; components reference a whole step.

```
// helix_design/tokens/semantic/type.json (excerpt) – composite type tokens
{
  "font": { "semantic": {
    "body": {
      "md": {
        "$description": "default body text",
        "size": { "$type": "dimension", "$value": "14px" },
        "lineHeight": { "$type": "number", "$value": 1.50 },
```



```

        "weight":      { "$type": "fontWeight", "$value":
"{font.primitive.weight.regular}" },
        "family":     { "$type": "fontFamily", "$value":
"{font.primitive.family.sans}" }
    },
    "title": {
        "lg": {
            "$description": "card title / ConnectButton label",
            "size":        { "$type": "dimension", "$value": "20px" },
            "lineHeight": { "$type": "number",     "$value": 1.30 },
            "weight":      { "$type": "fontWeight", "$value":
"{font.primitive.weight.semibold}" },
            "family":     { "$type": "fontFamily", "$value":
"{font.primitive.family.sans}" }
        },
        "mono": {
            "md": {
                "$description": "IP / pubkey / fingerprint",
                "size":        { "$type": "dimension", "$value": "13px" },
                "lineHeight": { "$type": "number",     "$value": 1.45 },
                "weight":      { "$type": "fontWeight", "$value":
"{font.primitive.weight.regular}" },
                "family":     { "$type": "fontFamily", "$value":
"{font.primitive.family.mono}" }
            }
        }
    } }
}

```

```

// helix_design/tokens/component/connect_button.json (excerpt) –
references the step
{
  "comp": { "connectButton": {
    "label": {
      "type": { "$type": "typeRef", "$value":
"{font.semantic.title.lg}" },
      "color": { "$type": "color", "$value":
"{color.semantic.text.onState}" }
    }
  } }
}

```

A component references a **whole** ramp step (`{font.semantic.title.lg}`) — it never re-specifies size/weight inline (a tier violation [DT §7]). The exporter expands the composite into each platform's native text-style object (§3.2).

3.2 Per-platform type emit (sample, all 6 forms)

The export pipeline [token-export-pipeline.md] expands one composite step into each target's idiomatic text style. The light/dark **colour** is applied by the component, not the type token (the type token is theme-invariant — size/weight/ family do not fork by theme; only the colour the text is painted in does [DT §5.1]).

```
/* CSS – dist/css/helix.css */
.hx-font-title-lg {
  font-family: var(--hx-font-family-sans);
  font-size: 20px; line-height: 1.30; font-weight: 600;
}

// Dart – dist/dart/helix_tokens.dart (TextStyle, colour applied
// by widget)
static const TextStyle titleLg = TextStyle(
  fontFamily: 'Inter', fontSize: 20, height: 1.30, fontWeight:
  FontWeight.w600);

// SwiftUI – dist/swift/HelixTokens.swift
static let titleLg = Font.custom("Inter", size:
  20).weight(.semibold)
// line-height applied via .lineSpacing(20 * (1.30 - 1.0)) at the
// call site

// Jetpack Compose – dist/compose/HelixTokens.kt
val TitleLg = TextStyle(
  fontFamily = InterFamily, fontSize = 20.sp, lineHeight = 26.sp,
  fontWeight = FontWeight.W600)

// ArkTS – dist/arkts/helix_tokens.ets
export const titleLg = { fontFamily: 'Inter', fontSize: 20,
  lineHeight: 26, fontWeight: 600 };

// C/Qt – dist/cqt/helix_tokens.h
#define HX_FONT_TITLE_LG_FAMILY "Inter"
#define HX_FONT_TITLE_LG_SIZE_PX 20
#define HX_FONT_TITLE_LG_WEIGHT 600
#define HX_FONT_TITLE_LG_LH_PX 26
```

UNVERIFIED (U-TIM-1). The exact line-height idiom per platform (CSS unitless multiplier vs. Compose lineHeight in sp vs. SwiftUI's .lineSpacing *additional* spacing) is pinned by the export-pipeline contract test [token-export-pipeline.md §drift-gate]. The samples above state the *intended* mapping (multiplier → absolute sp/px where the platform needs absolute); marked UNVERIFIED until that contract test exists (§11.4.6).

4. The no-overlap / no-collision typographic rule (§11.4.162)

§11.4.162 mandates "fonts MUST NOT collide, labels MUST NOT overlay labels". At the **type layer** this is concrete (layout enforcement is [component-library.md] + visual regression [visual-regression-and-qa.md]; this is the type half):

1. **Line-height ≥ 1.30 for any multi-line text block** so descenders of one line never touch ascenders of the next (collision). The ramp encodes this: every `body.* / title.*` step is ≥ 1.30 (`title.lg 1.30`, `body.* 1.45–1.50`); only single-line `label.*` (chips, never wrapping) and `display.*` (one line, generous) drop below.
2. **A label never overlaps another label.** Two text runs in one component (e.g. the `ExitPicker` row's primary `title.sm` + secondary `body.sm`) are separated by $\geq \text{space.scale.1}$ (4px) vertical gap [DT §6.1], and the `StatusChip` label has $\geq \text{space.scale.2}$ (8px) horizontal padding so the text never touches the chip edge or an adjacent chip.
3. **Truncation, never overflow.** Any text that may exceed its container (a long exit name, a wide pubkey) truncates with an ellipsis (proportional) or a middle-ellipsis (mono keys — preserve both ends) — it never overflows to overlay a neighbour. The full value is available on hover/long-press.
4. **One type family per role, no mixed-family runs.** A single text run is one family (sans **or** mono) — a key is rendered entirely `mono.*`, its label entirely sans — never a mid-string family switch that produces a baseline collision.

These four rules are asserted by the visual-regression suite (golden screenshots per theme, OCR-readback to confirm no clipped/overlapping glyphs, §11.4.162 / §11.4.168) in [visual-regression-and-qa.md]; this document supplies the type contract they assert against.

5. Iconography — set strategy, grid, asset forms

5.1 Set strategy — one cohesive line-icon set, brand-tinted

HelixVPN uses **one cohesive stroke (line) icon set** at a uniform 1.5px optical stroke, drawn on a 24×24 grid (§5.2). One set (not a mix of filled + line + brand logos from different families) is what keeps the UI cohesive — a §11.4.162 design-system requirement.

D-ICON-1 (surfaced, §10) — base icon set. The recommended base is a permissively-licensed open line-icon set (e.g. **Lucide**, ISC license) extended with HelixVPN-bespoke glyphs for the VPN-specific concepts no generic set has (shield-check, shield-slash, tunnel, relay-hop, kill-switch, exit-node, split-tunnel). Per §11.4.74 the base set is reused, not reinvented, and bespoke glyphs are added in the HelixVPN style. The **exact** base set + its license are **UNVERIFIED** pending the §11.4.99 verification pass (same discipline as fonts, §2.4) — Lucide/ISC is the *hypothesis*, confirmed before any SVG is committed. Composes [OV §13 D-DESIGN-5].

5.2 The sizing grid

Icons are authored on a **24×24** base grid with a **2px live-area padding** (a 20×20 optical safe area), so a glyph never touches the bounding box (the icon analogue of the §4.2 label-padding rule). Rendered sizes are a closed token set keyed to `size.*`:

Token	Size	Stroke	Use
<code>icon.size.xs</code>	12px	1.25px	inline-with-label.sm, dense badges
<code>icon.size.sm</code>	16px	1.5px	inline-with-body/label, list-row leading
<code>icon.size.md</code>	20px	1.5px	default — buttons, nav, StatusChip glyph
<code>icon.size.lg</code>	24px	1.5px	section headers, dialog icons
<code>icon.size.xl</code>	32px	2px	empty-state, the ConnectButton centre glyph
<code>icon.size.2xl</code>	48px	2.5px	onboarding / hero

Stroke scales with size so optical weight stays constant (a 12px icon at 1.5px looks heavier than a 32px icon at 1.5px — hence the per-size stroke).

5.3 The per-state status icon set (carries state WITHOUT colour — CVD a11y)

The single most product-specific icon group: each of the 7 `ffi::TunnelStatus` variants [FFI §3.2] has a **distinct glyph** so a colour-blind user reads the state from the *shape*, not only the [COLOR §3] colour (the [COLOR §0] honest boundary — colour is reinforcement, never the sole channel).

<code>ffi::TunnelStatus</code>	<code>icon.semantic.state.*</code>	Glyph	Why distinct shape
Disconnected	<code>state.disconnected</code>		

<code>ffi::TunnelStatus</code>	<code>icon.semantic.state.*</code>	Glyph	Why distinct shape
		shield (outline, hollow)	"off" — empty shield
Connecting	<code>state.connecting</code>	spinner / 3-dot (animated, §7)	"working"
Handshaking	<code>state.handshaking</code>	spinner + key (animated)	"working — authenticating"
Connected{direct}	<code>state.connected</code>	shield-check (filled)	"protected, P2P" — the success glyph
Connected{relay}	<code>state.connectedRelay</code>	shield-check + relay-hop dots	"protected, relayed"
Reconnecting	<code>state.reconnecting</code>	circular-arrows (pulsing, §7)	"retrying"
Down	<code>state.down</code>	shield-slash / broken-link	"dropped" — distinct from danger
Danger	<code>state.danger</code>	warning-triangle (!)	"exposed" — the universal alarm shape

The glyph + the [COLOR §3] colour + the text label are **three** redundant channels for the state — any one alone is sufficient to read it. Down (shield-slash) vs Danger (warning-triangle) are deliberately *different shapes*, not just different colours, so the two alarms are distinguishable under colour-vision deficiency (the [COLOR §3.2] rationale, satisfied at the icon layer).

5.4 Asset forms per platform

The icon source is **SVG** (the authoring form); the export pipeline [token-export-pipeline.md] emits the form each platform consumes:

Platform	Consumed icon form	How
Flutter (all app surfaces)	SVG via <code>flutter_svg</code> , or a generated icon font (.ttf + a Dart IconData map)	one source SVG → both; icon-font is smaller for the full set
Web (Console)	inline <code><svg></code> sprite (<code><symbol></code> + <code><use></code>) or an icon-font @font-face	sprite preferred (per-icon CSS colour)
iOS / macOS shim	SF Symbols where an equivalent exists; bespoke glyphs as SVG→PDF vector assets	system-consistent; bespoke as asset catalogue
Android / HarmonyOS shim	vector drawable (<code><vector></code>) / ArkTS resource	one source SVG → per-platform vector
Aurora (Qt)	SVG via Qt's SVG renderer	direct

Icons are **tinted at render** by a colour token (never baked-in colour in the SVG — the SVG uses `currentColor` / a single fill placeholder) so the same glyph serves light + dark (DS-I3) by swapping the tint token ([COLOR §3]); a state-icon's tint is its `color.semantic.state.*` value.

D-ICON-2 (surfaced, §10). SVG-direct vs. generated icon-font for Flutter (size vs. per-icon tint flexibility). Lean: **icon-font** for the static UI set (smaller, one glyph map) + **SVG-direct** for the few icons that need multi-colour or animation (the animated spinner/pulse, §7). Resolved at the Volume-10 assets review [OV §13]. UNVERIFIED until ratified.

6. Motion & animation token system

6.1 The motion token families

Motion lives in `motion.*` [DT §6.4]. Three leaf families compose into named transitions:

Family	\$type	Tokens
duration	duration	<code>motion.primitive.dur.{instant 0, fast 100, base 180, moderate 240, slow 360, pulse 1200}</code> (ms)
easing	cubicBezier	<code>motion.primitive.ease.{standard, decelerate, accelerate, emphasized, loop}</code>

Family	\$type	Tokens
named transition (semantic)	composite	motion.semantic.{press, stateXfade, sheet, connectPulse, ...} — each a {dur, ease} pair

6.2 The easing curves (concrete control points)

Token	Cubic-bézier	Character	Use
motion.primitive.ease.standard	cubic-bezier(0.2, 0, 0, 1)	quick-out, settle-in	most enter/move
motion.primitive.ease.decelerate	cubic-bezier(0, 0, 0, 1)	enters fast, eases to rest	elements entering the screen
motion.primitive.ease.accelerate	cubic-bezier(0.3, 0, 1, 1)	starts slow, exits fast	elements leaving the screen
motion.primitive.ease.emphasized	cubic-bezier(0.2, 0, 0, 1) (long dur)	expressive, for hero moments	connect success flourish
motion.primitive.ease.loop	cubic-bezier(0.4, 0, 0.6, 1)	symmetric in-out	the Connecting/Reconnecting pulse (loops without a visible seam)

6.3 The named transitions (semantic motion tokens)

Token	Duration	Easing	Use
motion.semantic.press	100ms (fast)	standard	button press scale/ink-ripple
motion.semantic.hover	100ms (fast)	standard	hover tint (desktop/web)
motion.semantic.stateXfade	180ms (base)	decelerate	cross-fade between two TunnelStatus colours+glyphs
motion.semantic.sheet		decelerate	

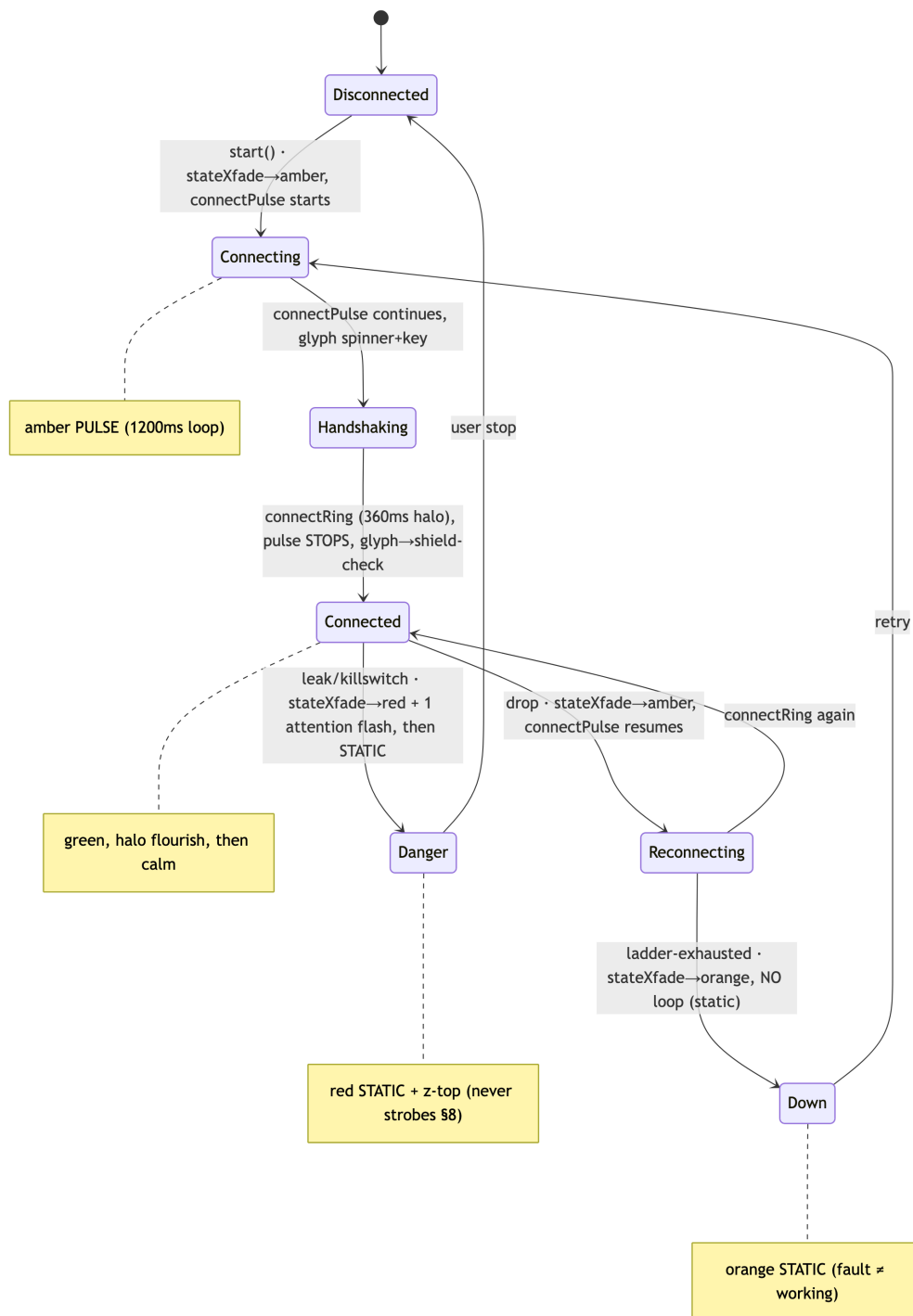
Token	Duration	Easing	Use
	240ms (moderate)		sheet / dialog enter
motion.semantic.sheetExit	180ms (base)	accelerate	sheet / dialog leave
motion.semantic.connectPulse	1200ms (pulse)	loop	the Connecting/ Reconnecting amber pulse loop
motion.semantic.connectRing	360ms (slow)	emphasized	the ConnectButton halo expand on reaching Connected
motion.semantic.toast	240ms in / 180ms out	decelerate/ accelerate	toast enter/ leave
motion.semantic.reducedMotion	0ms (instant)	—	the value every *.dur collapses to under OS reduce-motion (\$8)

```
// helix_design/tokens/semantic/motion.json (excerpt) – named-
transition composites
{
  "motion": { "semantic": {
    "stateXfade": { "dur":
{"$type":"duration","$value":"180ms"},
                    "ease":{"$type":"cubicBezier","$value":
[0.0,0,0,1]} },
    "connectPulse": { "dur":
{"$type":"duration","$value":"1200ms"},
                    "ease":{"$type":"cubicBezier","$value":
[0.4,0,0.6,1]} },
    "connectRing": { "dur":
{"$type":"duration","$value":"360ms"},
                    "ease":{"$type":"cubicBezier","$value":
[0.2,0,0,1]} }
  } }
}
```


7. The connection-state motion choreography (the FFI seam)

Motion is keyed to the 7-variant `ffi::TunnelStatus` [FFI §3.2] the same way colour ([COLOR §3]) and icons (§5.3) are — the three layers move together so a state transition reads as one coherent change, never three independent flickers.

Transition (FFI variant change)	Motion	What animates
any → any (colour/glyph change)	<code>stateXfade</code> (180ms)	the StatusChip/ ConnectButton fill and glyph and label cross-fade together (no in-between illegible frame, [COLOR §5.1])
→ Connecting / Handshaking	<code>connectPulse</code> (1200ms loop) starts	the amber fill breathes (opacity/scale loop); the spinner/3-dot glyph rotates
Connecting → Connected{*}	<code>connectRing</code> (360ms, emphasized)	a green halo expands once from the button; pulse stops, glyph swaps spinner→shield-check
→ Reconnecting	<code>connectPulse</code> (1200ms loop) resumes	amber breathes again; circular-arrows glyph rotates
→ Down	<code>stateXfade</code> to orange, no loop	shield-slash glyph settles (static — a fault is not "working")
→ Danger	<code>stateXfade</code> to red + a single <code>connectRing</code> -class attention flash, then static	warning-triangle settles on top of the z-danger layer [DT §6.5]; never strobes (§8)



The choreography rule. A *working* state (Connecting/Handshaking/Reconnecting) **loops** (pulse) — motion signals "in progress". A *settled* state (Connected/Down/Danger/Disconnected) is **static** after its entry transition — motion stops so a steady state looks steady. This is why Down and Danger get a one-shot entry (xfade / flash) but no loop: a leak banner that pulses forever would be both annoying and an all-y stroke hazard (§8). One source of truth for "is it moving?" = "is the tunnel still working on it?".

8. Reduced-motion & accessibility contract

§11.4.162 ally + the §11.4.107 no-flash discipline mandate that motion never harms a motion-sensitive or photosensitive user.

8.1 The reduce-motion substitution

When the platform reports **reduce-motion** (`UIAccessibility.isReduceMotionEnabled` / `Settings.Global.ANIMATOR_DURATION_SCALE==0` / CSS prefers-reduced-motion: reduce / equivalent), the export/runtime substitutes `motion.semantic.reducedMotion(0ms)` for every animation **duration**, with these state-specific degradations:

Normal motion	Reduce-motion degradation
<code>connectPulse</code> (amber breathing loop)	static amber — the Connecting/Reconnecting state shows a steady amber fill + a static spinner-glyph (or a textual "Connecting..."); it never pulses and never strobes
<code>stateXfade</code> (180ms cross-fade)	instant colour/glyph swap (0ms) — no fade, just the new state
<code>connectRing</code> (halo flourish)	omitted — the button simply <i>is</i> green; no expanding ring
Danger attention flash	omitted — the red banner appears statically (still z-top, still unmissable via colour+icon+text)

8.2 The hard no-strobe invariant

No HelixVPN animation flashes more than 3× per second, ever (WCAG 2.1 SC 2.3.1 — Three Flashes). The `connectPulse` at 1200ms is ~0.83 Hz (well under 3 Hz) even *without* reduce-motion; under reduce-motion every loop becomes static. There is no code path that produces a >3 Hz flash — the pulse period is a token (pulse 1200ms) the gate (§9 CM-motion-no-strobe) asserts is ≥ 334ms.

8.3 Motion never the sole channel

Like colour, motion is **reinforcement, never the only signal**. The pulse says "working", but the glyph (§5.3 spinner) and the text label ("Connecting...") say it too — a user with reduce-motion on, or who cannot perceive the pulse, still reads the state from the static glyph + label. The state is always conveyed by ≥ 2 non-motion channels (icon + text, plus colour) so removing motion removes **no** information (§11.4.107 / §11.4.137 content-correctness — the meaning survives the degradation).

9. Drift, contract & validation rules

Type/icon/motion tokens are single-source only if machine-checked. Gates run in `helix_design` CI + the consuming pre-build (paired §1.1 mutation each, [DT §9] pattern):

Gate	Asserts	Mutation that must FAIL it
CM-type-ramp-complete	every ramp step (§1.1) has all four of <code>size/lineHeight/weight/family</code> resolvable	delete the <code>weight</code> of <code>title.lg</code>
CM-type-lineheight-floor	every multi-line tier (<code>body.*</code> / <code>title.*</code>) has <code>lineHeight</code> ≥ 1.30 (§4.1 no-collision)	drop <code>body.md</code> line-height to 1.10
CM-type-mono-disambiguated	the mono family is on the disambiguation allow-list (§2.3)	swap mono to a family with ambiguous <code>0/0</code>
CM-icon-state-coverage	all 7 <code>ffi::TunnelStatus</code> variants (§5.3) have a distinct <code>icon.semantic.state.* glyph</code>	remove <code>state.down</code> (would collide visually with <code>danger</code>)
CM-icon-grid-padding	every source SVG keeps ≥ 2 px live-area padding on the 24-grid (§5.2)	author a glyph touching the bounding box
CM-motion-no-strobe	every loop/pulse duration ≥ 334 ms (≤ 3 Hz, §8.2)	set <code>connectPulse.dur</code> to 200ms (5 Hz)
CM-motion-reduced-complete	every animated named transition (§6.3) has a defined reduce-motion degradation (§8.1)	add a new transition with no reduce-motion mapping
CM-motion-state-coverage	every <code>TunnelStatus</code> transition (§7) has a defined motion	leave <code>→ Danger</code> with no motion mapping
CM-tim-no-drift	every emitted form (CSS/Dart/Swift/Compose/ArkTS/C-Qt) of type/icon/motion regenerates byte-identically from the JSON source; a hand-edit is a finding [token-export-pipeline.md]	hand-edit <code>HelixTokens.kt TitleLg</code>

The drift gate (CM-tim-no-drift) is the load-bearing one — same role as [DT §9 CM-token-no-drift]: the only way a platform's type/icon/motion can diverge from source is a hand-edit, and the gate catches it. Exported docs of these tables carry the §11.4.65/.168 visual-validation per [visual-regression-and-qa.md].

10. Surfaced decisions & cross-doc contracts

id	Decision / contract	Status
D-TYPE-1	Native shims (iOS NE / Android tile / HarmonyOS ability / Aurora Qt) use the system font , not bundled brand sans (binary-size vs. minimal-text trade-off); the Flutter app is always brand-font.	recommended, surfaced
D-TYPE-2 UNVERIFIED	Font choice Inter (sans) + JetBrains Mono (mono) + their licenses (OFL/ Apache assumed) MUST be verified against latest upstream per §11.4.99 before any font file is committed; OFL/ Apache is the hypothesis, not asserted fact.	open — §11.4.99 pass
D-ICON-1 UNVERIFIED	Base icon set = a permissive open line set (Lucide/ISC hypothesised) + HelixVPN-bespoke VPN glyphs (shield-check/slash, tunnel, relay, kill-switch); license verified per §11.4.99 before any SVG committed.	open — §11.4.99 pass
D-ICON-2	Flutter icon delivery: icon-font for the static set + SVG-direct for animated/ multi-colour glyphs.	recommended, surfaced
D-TIM-1	Type ramp anchored at body.md=14px, ratio 1.25, snapped to the 4-pt grid; body.lg=16px for reading prose.	decided
D-TIM-2	Motion choreography: <i>working</i> states loop (pulse), <i>settled</i> states are static after their one-shot entry transition (§7).	decided
C-TIM-A (consumes)	The 7-variant ffi::TunnelStatus vocabulary owned by [FFI §3.2]; §5.3 (icons) + §7 (motion) stay total over it (gates CM-icon-state-coverage, CM-motion-state-coverage). A new FFI variant is a contract change here.	contract

id	Decision / contract	Status
C-TIM-B (consumes)	The token tier/naming model + the font/motion \$type envelopes from [design-tokens.md §2/§3/§6.4]; the colours type/icons paint in from [color-system.md §2/§3].	contract
C-TIM-C (provides)	The type/icon/motion token shapes (§3, §5.4, §6) are the input contract for [token-export-pipeline.md]'s type/icon/motion emit.	contract
C-TIM-D (provides)	The §4 / §8 type-collision + reduce-motion contracts are the assertions [visual-regression-and-qa.md] re-runs as automated gates (§11.4.162 / §11.4.107).	contract
U-TIM-1 UNVERIFIED	The exact per-platform line-height idiom (multiplier vs. absolute sp/px) + the ArkTS/C-Qt type-emit shapes — pinned by the export-pipeline contract test; UNVERIFIED until it exists.	open

Sources verified

- **Type ramp (tiers, sizes, line-heights, weights, the modular scale + 14px anchor), font-family strategy + fallback chain, icon set/grid/asset-form policy, the per-state status-icon set, the motion durations/curves/named transitions, the connection-state choreography, the reduce-motion + no-strobe contract — NO external source needed — original HelixVPN design work**, modeled on the well-established modular-type-scale, 24-grid line-icon, and Material/M3-class motion-token conventions (engine-neutral; the authoring engine is OpenDesign per §11.4.162, [opendesign-foundation.md] §3.2 DESIGN.md §3 Typography / §6 Motion).
- **7-variant ffi::TunnelStatus vocabulary (Disconnected · Connecting · Handshaking · Connected{direct|relay} · Reconnecting · Down · Danger)** — final/v04-client/ffi-surface.md §3.2–§3.3 (read 2026-06-25); the icon (§5.3)
 - motion (§7) sets are held total over it by gates CM-icon-state-coverage / CM-motion-state-coverage.
- **Token tier/naming model + font.*/motion.* \$type envelopes the type/ motion tokens live in; the elevation/colour tokens icons are tinted by** — final/v10-design/design-tokens.md §1/§2/§3/§6.4 +

final/v10-design/color-system.md §2/§3 (siblings, this wave, read 2026-06-25).

- **3-app / 8-platform matrix, the per-platform font/icon/native-shim languages (SwiftUI / Compose / ArkTS / C-Qt), the OpenDesign DESIGN.md authoring form** — final/SPECIFICATION.md §3 + final/v10-design/00-overview-and-submodule.md §6 + final/v10-design/opendesign-foundation.md §3.2 (read 2026-06-25).
- **WCAG SC 2.3.1 (Three Flashes, the ≤3 Hz no-strobe floor) + the reduce-motion platform flags (prefers-reduced-motion, isReduceMotionEnabled, ANIMATOR_DURATION_SCALE)** — *W3C Web Content Accessibility Guidelines 2.1* SC 2.3.1 / SC 2.3.3 (<https://www.w3.org/TR/WCAG21/#three-flashes-or-below-threshold>, verified 2026-06-25) + the named platform reduce-motion APIs (Apple / Android / CSS) — the API names are platform-standard; the §8.1 substitution policy is HelixVPN's own design.
- **Font + icon-set licensing (Inter, JetBrains Mono, Lucide)** — UNVERIFIED (D-TYPE-2 / D-ICON-1): the OFL/Apache/ISC assumptions are **NOT** asserted as fact; they are the hypotheses the mandatory §11.4.99 latest-source verification pass confirms (with the license file + verified URL + access date recorded) **before** any font/icon file is committed. Stated as unverified per §11.4.6.
- Items explicitly marked UNVERIFIED (U-TIM-1, D-TYPE-2, D-ICON-1) are pending their named contract test / verification pass per §11.4.6 — not asserted as fact.